

Resource-efficient Deep Neural Networks for Automotive Radar Interference Mitigation

Johanna Rock, Wolfgang Roth, Mate Toth, Paul Meissner, and Franz Pernkopf

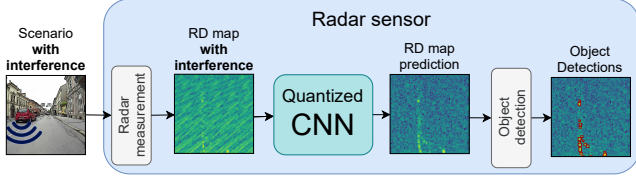


Figure 1: Interference mitigation of radar signals using a quantized CNN to remove interference patterns, retain object signals, and provide a high detection sensitivity. To use this approach in practice, a resource-efficient model is essential.

Abstract—Radar sensors are crucial for environment perception of driver assistance systems as well as autonomous vehicles. With a rising number of radar sensors and the so far unregulated automotive radar frequency band, mutual interference is inevitable and must be dealt with. Algorithms and models operating on radar data are required to run the early processing steps on specialized radar sensor hardware. This specialized hardware typically has strict resource-constraints, i.e. a low memory capacity and low computational power. Convolutional Neural Network (CNN)-based approaches for denoising and interference mitigation yield promising results for radar processing in terms of performance. Regarding resource-constraints, however, CNNs typically exceed the hardware’s capacities by far.

In this paper we investigate quantization techniques for CNN-based denoising and interference mitigation of radar signals. We analyze the quantization of (i) weights and (ii) activations of different CNN-based model architectures. This quantization results in reduced memory requirements for model storage and during inference. We compare models with fixed and learned bit-widths and contrast two different methodologies for training quantized CNNs, i.e. the straight-through gradient estimator and training distributions over discrete weights. We illustrate the importance of structurally small real-valued base models for quantization and show that learned bit-widths yield the smallest models. We achieve a memory reduction of around 80% compared to the real-valued baseline. Due to practical reasons, however, we recommend the use of 8 bits for weights and activations, which results in models that require only 0.2 megabytes of memory.

Index Terms—Quantization aware training, resource-efficiency, binarized convolutional neural networks, straight-through estimator, discrete weight distributions, uncertainty maps, interference mitigation, automotive radar.

I. INTRODUCTION

Johanna Rock, Wolfgang Roth and Franz Pernkopf are with the Signal Processing and Speech Communication Laboratory, Graz University of Technology, Austria, e-mail: {johanna.rock, roth, pernkopf}@tugraz.at.

Mate Toth and Paul Meissner are with Infineon Technologies Austria AG, Graz, e-mail: {mate.toth, paul.meissner}@infineon.com.

ADVANCED Driver Assistance Systems (ADAS) and Autonomous Vehicles (AV) heavily rely on a multitude of heterogeneous sensors for environment perception. Among them are radar sensors that are used for object detection, classification and to directly measure relative object velocities. Advantages of radar sensors are a high range resolution and their robustness concerning difficult weather and lighting conditions.

Typically *frequency modulated continuous wave (FMCW)/chirp sequence (CS)* radars are used in the automotive context. They transmit sequences of linearly modulated radio frequency (RF) chirps in a shared and non-regulated band. This may lead to mutual interference of multiple radar sensors; it becomes increasingly likely with a higher number of deployed radar-enhanced vehicles and larger chirp bandwidths of individual sensors used for better range-resolution.

For a non-regulated spectrum, the most common form of mutual interference is non-coherent, where the transmitters send with non-identical parameters. This results in burst-like interferences in time domain and a decreased detection sensitivity in the range-Doppler (RD) map [1], [2]. Thus, the detection and mitigation of interference is crucial in a safety context and must be addressed.

Several conventional signal processing algorithms for interference mitigation of mutual interference have been proposed. The most simplistic method is to substitute all interference-affected samples with zero [3], followed by an optional smoothing of the boundaries. More advanced methods use nonlinear filtering in slow-time [4], iterative reconstruction using Fourier transforms and thresholding [5], estimation and subtraction of the interference component [6], an adaptive noise canceller [7], or beamforming [8].

Recently, the use of deep learning has emerged for radar spectra denoising and interference mitigation. For this task deep neural networks (DNNs) are applied in time domain or in frequency domain, typically in a supervised manner. For interference mitigation in time domain, recurrent neural networks (RNNs) are used in [9], [10]. For interference mitigation in frequency domain, CNN-based models [11], [12], Convolutional Autoencoders [13] and U-Net inspired CNNs [14] are used.

While the results are impressive on simulated and real-world measurement data, the problem of high memory and computational requirements of DNN models has not been addressed in sufficient detail so far. In order to use the aforementioned methods for interference mitigation in practice, they have to comply with memory, computational as well as real-time